

Sri Lanka Institute of Information Technology



IT3060

Human Computer Interaction

Lab 5- Activity 3

Technology Comparison Matrix

IT23657878- W.R.S.D.Diwyanjali

1. Objective

The objective of this activity is to consolidate the frontend, backend, database and authentication comparisons from Activities 1 and 2 into a weighted technology decision matrix. The matrix evaluates the technologies against FitFlow's project needs, including performance, scalability, development speed, security, cost, AI/ML support and maintainability.

2. FitFlow Decision Criteria and Weights

Criterion	Weight	Reason for FitFlow
Performance	20%	FitFlow requires responsive fitness interactions, animations and fast application performance.
Scalability	15%	The redesigned application should support growing users, social activity and notifications.
Development Speed	15%	The mid-sized startup needs efficient development and faster delivery.
Security & Privacy	15%	The application handles wellness-related information and requires strong data protection.
AI/ML Support	10%	AI-powered personalized workouts and computer-vision nutrition features are core requirements.
Real-time Capability	10%	Social interactions and scalable notifications require real-time capabilities.
Cost	5%	The solution should be financially manageable for a mid-sized startup.
Maintainability	10%	The selected stack should be manageable by the development team over time.

Scoring scale: 1 = Poor, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent. Weighted score = $(\text{Score} \div 5) \times \text{Weight}$.

3. Frontend Technology Decision Matrix

Technology	Performance	Scalability	Dev. Speed	Security	AI/ML	Real-time	Cost	Maintainability
Flutter	4	4	5	4	4	4	4	4
React Native	5	4	5	4	5	5	4	5
Kotlin Multiplatform	5	4	3	5	4	4	3	4
Swift/SwiftUI	5	2	3	5	5	4	2	4

4. Backend Technology Decision Matrix

Technology	Performance	Scalability	Dev. Speed	Security	AI/ML	Real-time	Cost	Maintainability
Node.js / Express	4	5	5	4	4	5	5	5
NestJS	4	5	4	4	4	5	4	5
Python / FastAPI	4	5	4	4	5	4	4	5
Go	5	5	3	5	4	5	4	4

5. Database Technology Decision Matrix

Technology	Performance	Scalability	Dev. Speed	Security	AI/ML	Real-time	Cost	Maintainability
PostgreSQL	4	5	4	5	4	4	4	5
MongoDB	4	5	5	4	4	4	4	4
Firebase / Firestore	4	5	5	4	4	5	3	4
DynamoDB	5	5	3	5	4	4	3	4

6. Authentication Technology Decision Matrix

Technology	Performance	Scalability	Dev. Speed	Security	AI/ML	Real-time	Cost	Maintainability
Firebase Authentication	4	5	5	5	4	5	4	5
AWS Cognito	4	5	3	5	4	4	4	4
Auth0	4	5	4	5	4	4	3	5
Supabase Auth	4	4	5	4	4	4	4	4

7. Weighted Decision Matrix – Recommended Stack

Technology	Perf.	Scale	Speed	Security	AI/ML	Real-time	Cost	Maintain.	Weighted Total
React Native	5	4	5	4	5	5	4	5	93.0%
Node.js / Express	4	5	5	4	4	5	5	5	91.0%
Firebase / Firestore	4	5	5	4	4	5	3	4	87.0%
Firebase Authentication	4	5	5	5	4	5	4	5	93.0%

The weighted totals are calculated using the FitFlow-specific criteria and weights above. The matrix is used to support the overall stack decision rather than to claim that one technology is universally better for all projects.

8. Recommended Technology Stack

Based on the weighted evaluation and the supplied FitFlow case study, the recommended stack is:

Frontend: React Native

Backend: Node.js with Express

Database / real-time data: Firebase Firestore

Authentication: Firebase Authentication

9. Supporting Rationale

React Native provides a shared mobile codebase and rapid development for iOS and Android, matching the cross-platform direction of FitFlow.

Node.js with Express supports rapid API development and integrates naturally with the JavaScript/React ecosystem.

Firebase/Firestore supports real-time social functionality and scalable notifications, which are explicitly required in the FitFlow scenario.

Firebase Authentication provides an integrated authentication option that reduces the number of separate services the team must maintain.

The combination supports AI services, camera-based nutrition features, notifications and real-time interactions through appropriate APIs, SDKs and integrations.

The selected combination is consistent with the technology stack described in the supplied FitFlow case study.

10. Conclusion

The technology comparison matrix converts the findings from Activities 1 and 2 into a structured decision. The selected weights reflect FitFlow's need for performance, scalability, rapid development, security, AI/ML support, real-time functionality, manageable cost and maintainability. The resulting recommendation is React Native for the frontend, Node.js with Express for the backend, Firebase/Firestore for real-time application data, and Firebase Authentication for user authentication.

11. Source Basis

This activity is based on the supplied IT3060 Lab Exercise 05 requirements and the FitFlow Redesign Case Study. The case study specifically describes React Native for the frontend and Node.js with Express and Firebase for backend and real-time functionality.